

# 13.75 BILLION YEARS THE AGE OF OUR UNIVERSE

This super computer simulation of the structure of the Universe from 2005, created at the Max Planck Institute in Germany, shows the distribution of dark matter in part of the Universe.

**IN THIS YEAR, THE CHINESE NATIONAL SPACE** administration launched its first manned space mission, Shenzhou 5. Astronaut Yang Liwei spent 21 hours and 23 minutes in space, orbiting Earth 14 times.

In February, the US **space shuttle Columbia disintegrated while it was reentering** Earth's atmosphere. The incident resulted in the halting of the space shuttle program for two years—almost

as long as the gap in the program caused by the loss of the *Challenger* (see 1986).

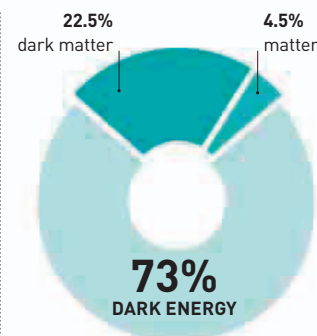
The same month, cosmologists and astrophysicists published the first year's observations of the **Wilkinson Microwave Anisotropy Probe (WMAP)**. One of the aims of the program was to provide a more detailed map of the cosmic background radiation than the COBE satellite did (see 1992). Variations in the cosmic

background radiation (CMB) reflect variations in the density of the early Universe, which in turn caused matter to clump together, forming galaxies and galaxy clusters. The map produced by WMAP also refined an emerging

**Yang Liwei in space**  
*Chinese astronaut, Yang Liwei, is shown aboard the Shenzhou 5 capsule. China became only the third nation to send humans into space, joining the US and Russia.*

model of the Universe, the Lambda-Cold Dark Matter model (see p.345). The lambda part is the **cosmological constant**—a repulsive force that is accelerating the Universe's expansion. Evidence for the existence of the cosmological constant—or dark energy—came from studies of supernovas in other galaxies (see 1998). Cold dark matter is matter that neither produces nor interacts with electromagnetic radiation. The presence of cold dark matter can be inferred only by its gravitational interaction with ordinary matter. WMAP's observations also enabled scientists to put the Universe's age, with a high degree of accuracy, at **13.7 billion years** (it has since been updated to closer to 13.82 billion years).

Three years after the draft version of the human genome was published, the International Human Genome Sequencing Consortium finally **completed the full sequencing of all 3**



**Composition of the Universe**  
According to WMAP observations, ordinary matter accounts for only a small percentage of the Universe's total mass-energy.

**billion DNA base pairs** that make up the human genome. A draft of the chimpanzee genome, which is nearly 99 percent identical to the human genome, was also published in 2003. The chimpanzee is the closest living relative to humans, and comparisons between the two genomes give biologists an unprecedented opportunity to study primate evolution.

**“ I’M FEELING VERY GOOD IN SPACE, AND IT LOOKS EXTREMELY SPLENDID AROUND HERE. ”**

**Yang Liwei, Chinese astronaut,** on the telephone to his wife from space, 2003



**February 11** Wilkinson Microwave Anisotropy Probe (WMAP) maps cosmic background radiation

**February 1** Space shuttle Columbia disaster kills all seven astronauts on board

**March** Footprints found in Italy, dated to 350,000 years ago, are the oldest known footprints of direct human ancestors

**April 14** The successful completion of the Human Genome Project is announced

**October 15** Yang Liwei is China's first astronaut, aboard the Shenzhou 5 mission

**December 12** The first draft sequence of the chimpanzee genome is published

**December 20** The European Space Agency's first Mars probe, Mars Express, reaches Mars